StreetCare AI — deployment architecture

One Docker image, one container, two processes: Streamlit is the only published port; the REST API, its stores and the models live inside the container

Built from Project/Dockerfile · docker-entrypoint.sh · .github/workflows/build.yml · config/config.py

1 Access & network

Staff roles
three roles, a browser and a sign-in

- **Street Care Coordinator** — queue · opens reviews
- **Highways Inspector (2 / district)** — decides reviews
- **Customer Services Adviser** — status & chat only

uses

Web browser
any modern browser

`http://<host>:8080`

Streamlit pages over a websocket
token kept server-side in
`st.session_state`

:8080

Host or deployment platform
Docker engine · the platform may set PORT

```
docker build -t streetcare-ai .
docker run --rm -p 8080:8080 \
    --env-file .env -v streetcare-db:/app/database \
    streetcare-ai
```

README command + the two flags it needs

-p 8080:8080 → container :8080 (the only published port)

Google Gemini API
egress

external LLM · HTTPS via google-genai

model = GEMINI_MODEL from the env file
review start: 5 calls (one per agent)
decision: +2 ORDER · +1 INSPECT · +0 REFER
chat: at most 1 call per question

HTTPS · GEMINI_API_KEY

Hugging Face Hub
egress

embedding model download

`all-MiniLM-L6-v2`
fetched on the first embedding call
(knowledge search) unless already
in the container's model cache

first embedding call

2 Runtime — one container

Docker container `streetcare-ai` — `python:3.11-slim · WORKDIR /app · EXPOSE 8080 · ENTRYPOINT ./docker-entrypoint.sh`

Process 1 · Streamlit dashboard
`run_dashboard.py · 0.0.0.0:${PORT:-8080} · PID 1`

- pings the API's /docs first; if it does not answer shows 'The service is not answering…'
- sign-in page, then only the role's pages:
  Report queue · Report detail · Decision queue
  Audit trail · Street Care chat
- `dashboard/api_client.py` — the only HTTP client
- `.streamlit/config.toml`: theme, minimal toolbar, usage statistics off · holds no business rule

exec

docker-entrypoint.sh
sh · set -eu — the container's start command

```
1  python run_api_service.py &
     api_pid=$!   (background child)
2  exec streamlit run run_dashboard.py \
     --server.port=${PORT:-8080}
   trap INT TERM EXIT → kill $api_pid
```

starts

httpx → localhost:8081

X-Session-Token
timeout 300 s

Process 2 · FastAPI REST service
`run_api_service.py → uvicorn 0.0.0.0:8081 · not published`

Start-up gate
13 required files must exist:
data, processed CSVs, 3 models,
2 schemas, index, streetcare.db

Warm-up at import
build_workflow(checkpoint.db)
clean + screen 40 open reports once
one lock per report (in memory)

Guard · api/ + access/
session token → role → district
401 · 403 · 404 · 409 · 422 · 502

11 endpoints + OpenAPI
/auth · /queue · /reviews · /chat · /audit
docs at :8081/docs (internal)

LangGraph review workflow
10 nodes · 7 agents · interrupt() gate
SqliteSaver → checkpoint.db
survives a container restart*

Tools + 3 scikit-learn models
joblib .pkl loaded once, in memory
screening · register · evidence
likelihood · crew hours · sentences

Knowledge search
ChromaDB PersistentClient
all-MiniLM-L6-v2 on CPU
standards parsed at run time

Language-model gateway
agentic/language_model.py
temperature 0.2 · ≤ 8 192 tokens out
every call logged to llm_calls

R/W

Writable state `/app/database`
mount a volume — --rm deletes it otherwise

`streetcare.db`
SQLite · 6 tables:
users, runs, reviews,
decisions, llm_calls, chat

`checkpoint.db`
LangGraph SqliteSaver
paused review runs
WAL mode

`chroma_store/`
chroma.sqlite3 · 205 passages from 16 PDFs

read

Read-only artefacts
baked in by COPY . . at build time

`data/`	8 CSVs + 16 PDFs · 17 MB
`processed/`	5 clean CSVs · 17 MB
`output/models/`	3 .pkl + 2 schemas · 13 MB
`config/config.py`	every setting, no secrets

Ports and processes

`:8080`	**Streamlit (PID 1)**	published with -p
`:8081`	**FastAPI / uvicorn**	inside the container
`:8081/docs`	**OpenAPI page**	inside the container
`local run`	**same two ports**	two terminals, myenv

Secrets and settings
`.env`
GEMINI_API_KEY · GEMINI_MODEL · FRONTEND_PORT
`.dockerignore`
drops .env (and venvs, caches) from the image —
pass it at run time with --env-file
`config/config.py` holds every other value: ports, paths, limits

Start-up order
1. entrypoint starts the API in the background
2. API checks the 13 required files (exit 1 if any is missing)
3. imports routes: workflow, models, index; screens 40 reports
4. uvicorn listens on :8081
5. Streamlit serves :8080 — first page pings /docs

image streetcare-ai → docker run

3 Build & delivery

Team member
5 branches, one per
teammate · git push
or pull request

push

.github/workflows/build.yml
on: push to any branch · pull_request

```
1  actions/checkout@v2  (fetch-depth: 0)
2  SonarCloud scan  → SonarQube server
3  Trigger API  (if: always())
     POST {repo_id, branch_name, org_name}
```

runs on ubuntu-latest · quality scan only:
it builds no image and runs no tests

SonarQube
sonar.server.examly.io

code-quality dashboard
per branch

Assessment platform
API_URL from repo secrets

told which repo and branch
was pushed

Offline build — run once from Project/
produces every artefact the image copies

```
1  python run_data_pipeline.py
     → processed/*.csv · data_quality_report.json
2  python run_model_training.py
     → output/models/ · model_metrics.json
3  python run_knowledge_indexing.py → chroma_store/
4  python run_database_setup.py  → streetcare.db
5  pytest tests.py               → 50 tests
```

files

docker build -t streetcare-ai .
image recipe — Project/Dockerfile

```
FROM python:3.11-slim
pip install -r requirements.txt
  + CPU-only torch wheel index
COPY . .
  code + data + processed + models + index
RUN chmod +x docker-entrypoint.sh
EXPOSE 8080 · ENTRYPOINT the script
```

Deployment notes

Single replica. SQLite files and the per-report locks live in one process, so run exactly one container; scale up, not out.

Only :8080 is public. The REST API and /docs answer only inside the container; the dashboard is the sole way in.

Embedding package. requirements.txt does not list sentence-transformers, which the index's embedding function needs.

Long requests. Starting a review waits for 5 Gemini calls; the dashboard allows 300 s per request.

Keep /app/database. With --rm and no volume, reviews, decisions, checkpoints, chat audit and sessions vanish on stop.

Process supervision. Streamlit is PID 1; if the background API exits, the page shows 'service not answering'.

Offline hosts. Pre-download all-MiniLM-L6-v2 into the image; otherwise the first search needs Hugging Face.

CI scope. build.yml scans code quality only; the 50 tests run locally with pytest tests.py.

** a paused review resumes after a restart only if checkpoint.db is on the mounted volume.*

Human actor · User interface · Deterministic code / config · LLM-backed component · Trained ML model · Data store / external service · Gate / secrets · request / data flow · outbound call to an external service